ASSIGNMENT 12

Suggested Solutions

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CHAPTER 10

6. Prove Theorem 10.19.

Solution:

Let $\mathbf{Y} = A\mathbf{X}$. Then

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{|\det(A)|} f_{\mathbf{X}}(A^{-1}\mathbf{y}).$$

Letting $\mathbf{x} = A^{-1}\mathbf{y}$ in (10.50), we have

$$\begin{aligned} h(\mathbf{X}) &= -\int_{S_{\mathbf{X}}} f_{\mathbf{X}}(\mathbf{x}) \log f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} \\ &= -\int_{S_{\mathbf{Y}}} \frac{1}{|\det(A)|} f_{\mathbf{X}}(A^{-1}\mathbf{y}) \log f_{\mathbf{X}}(A^{-1}\mathbf{y}) d\mathbf{y}. \end{aligned}$$

The rest are the same as in the proof of Theorem 10.15.

7. For continuous random variables X and Y, discuss why $I(X;X)$ is not equal to $h(X)$.

Solution:

For X be a continuous random variable such that $h(X) < \infty$. Consider

$$
\begin{aligned}
H(\hat{X}_\Delta) &= -\sum_i \Pr\{\hat{X}_\Delta = i\} \log \Pr\{\hat{X}_\Delta = i\} \\
&\approx \sum_i f(x_i)\Delta \log(f(x_i)\Delta) \\
&= -\sum_i f(x_i)\Delta(\log f(x_i) + \log \Delta) \\
&= -\sum_i (f(x_i) \log f(x_i))\Delta - (\log \Delta) \sum_i f(x_i)\Delta \\
&\approx h(X) - (\log \Delta) \int f(x)dx \\
&= h(X) - \log \Delta.
\end{aligned}
$$

Therefore, $H(\hat{X}_\Delta) \to \infty$ as $\Delta \to 0$. Now consider

$$
I(X;X) \approx I(\hat{X}_\Delta; \hat{X}_\Delta) = H(\hat{X}_\Delta) \to \infty
$$

as $\Delta \to 0$. This shows that $I(X;X)$ cannot be the same as $h(X)$.

8. Each of the following continuous distributions can be obtained as the distribution that maximizes the differential entropy subject to a suitable set of constraints:

   (a) the exponential distribution,

   $$f(x) = \lambda e^{-\lambda x}$$

   for $x \geq 0$, where $\lambda > 0$;

   (b) the Laplace distribution,

   $$f(x) = \frac{1}{2}\lambda e^{-\lambda |x|}$$

   for $-\infty < x < \infty$, where $\lambda > 0$;

No need to answer (c),(d)&(e) for Assignment 12.

The solutions are included for your self-study.

   (c) the gamma distribution,

   $$f(x) = \frac{\lambda}{\Gamma(\alpha)}(\lambda x)^{\alpha-1}e^{-\lambda x}$$

   for $x \geq 0$, where $\lambda, \alpha > 0$ and $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$;

   (d) the beta distribution,

   $$f(x) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)}x^{p-1}(1-x)^{q-1}$$

   for $0 \leq x \leq 1$, where $p, q > 0$;

   (e) the Cauchy distribution,

   $$f(x) = \frac{1}{\pi(1+x^2)}$$

   for $-\infty < x < \infty$.

   Identify the corresponding set of constraints for each of these distributions.

   Solution:

   The solutions are obtained through applications of Corollary 10.42, which can be rephrased as follows. Let f^* be any pdf defined on $\Re^n$ and suppose for all $\mathbf{x} \in \mathcal{S}_{f^*}$

   $$\log f^*(\mathbf{x}) = -\lambda_0 - \sum_{i=1}^{m} \lambda_i r_i(\mathbf{x}),$$

or

$$f^*(\mathbf{x}) = e^{\log f(\mathbf{x})} = e^{-\lambda_0 - \sum_{i=1}^m \lambda_i r_i(\mathbf{x})}.$$

Let $\mathcal{S} = \mathcal{S}_{f^*}$. Then Corollary 10.42 says that $f^*(\mathbf{x})$ maximizes $h(f)$ over all pdf f defined on $\mathcal{S}_{f^*}$, subject to the constraints

$$E_f[r_i(\mathbf{X})] = E_{f^*}[r_i(\mathbf{X})]$$

for $1 \le i \le m$.

We now give the solution to Part (c), the gamma distribution. For $x \ge 0$, write

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}.$$

Then

$$\log f(x) \quad = \quad \log \frac{\lambda^\alpha}{\Gamma(\alpha)} + (\alpha - 1)\log x - \lambda x.$$

By Corollary 10.42, $f(x)$ maximizes $h(f)$ over all pdf g defined on $\Re^+$, subject to the constraints

$$E_g[\log X] = E_f[\log X]$$

and

$$E_g[X] = E_f[X].$$

9. Let μ be the mean of a continuous random variable X defined on $\Re^+$. Obtain an upper bound on $h(X)$ in terms of μ.

Solution:

Let Y be exponentially distributed with parameter λ, i.e.,

$$f(y) = \lambda e^{-\lambda y}, \quad y \geq 0.$$

Then $EY = \mu = \lambda^{-1}$. Letting e as the base of the logarithm, we have

$$\begin{aligned} h(Y) &= -\int_{y=0}^{\infty} f(y) \ln f(y) dy \\ &= -\int_{y=0}^{\infty} f(y)(\ln \lambda - \lambda y) dy \\ &= -\ln \lambda + \lambda \mu \\ &= \ln \mu + 1. \end{aligned}$$

Then by Theorem 10.41, for any continuous random variable X defined on $\Re^+$, we have

$$h(X) \leq h(Y) = \ln \mu + 1.$$

For a general base of the logarithm, we have

$$h(X) \leq \log \mu + \log e.$$

12. *Hadamard's inequality* Show that for a positive semidefinite matrix K, $|K| \leq \prod_{i=1}^{n} K_{ii}$, with equality if and only if K is diagonal. Hint: Consider the differential entropy of a multivariate Gaussian distribution.

Solution:

Let $\mathbf{X} \sim \mathcal{N}(0, K)$. Then

$$h(\mathbf{X}) = \frac{1}{2} \log(2\pi e|K|)$$

and

$$h(X_i) = \frac{1}{2} \log(2\pi e K_{ii}).$$

Then substitute the above into the inequality

$$h(\mathbf{X}) \leq \sum_{i} h(X_i)$$

to obtain the desired result.

13. Let $K_{\mathbf{X}}$ and $\tilde{K}_{\mathbf{X}}$ be the covariance matrix and the correlation matrix of a random vector $\mathbf{X}$, respectively. Show that $|K_{\mathbf{X}}| \leq |\tilde{K}_{\mathbf{X}}|$. This is a generalization of $\text{var}X \leq EX^2$ for a random variable X. Hint: Consider a multivariate Gaussian distribution with another multivariate Gaussian distribution with zero mean and the same correlation matrix.

Solution:

Let $\mathbf{X} \sim \mathcal{N}(\mathbf{m}, K_X)$ and $\mathbf{X}' \sim \mathcal{N}(0, \tilde{K}_X)$. Since $\mathbf{X}'$ is zero mean, $\tilde{K}_{\mathbf{X}'} = K_{\mathbf{X}'} = \tilde{K}_{\mathbf{X}}$, i.e., $\mathbf{X}$ and $\mathbf{X}'$ have the same correlation matrix. Then by Theorem 10.45, we have

$$\frac{1}{2}\log(2\pi e|K_{\mathbf{X}}|) = h(\mathbf{X}) \leq h(\mathbf{X}') = \frac{1}{2}\log(2\pi e|K_{\mathbf{X}'}|) = \frac{1}{2}\log(2\pi e|\tilde{K}_{\mathbf{X}}|),$$

or $|K_{\mathbf{X}}| \leq |\tilde{K}_{\mathbf{X}}|$.